

| Tool Version : Vivado v.2025.2 (win64) Build 6299465 Fri Nov 14
19:35:11 GMT 2025
| Date : Sat Jul 18 14:22:46 2026
| Host : AK_LAPTOP running 64-bit major release (build
9200)
| Command : report_utilization -file
cpu_top_utilization_synth.rpt -pb cpu_top_utilization_synth.pb
| Design : cpu_top
| Device : xc7a15tcp236-1
| Speed File : -1
Design State : Synthesized

Utilization Design Information

Table of Contents

-
1. Slice Logic
 - 1.1 Summary of Registers by Type
 2. Memory
 3. DSP
 4. IO and GT Specific
 5. Clocking
 6. Specific Feature
 7. Primitives
 8. Black Boxes
 9. Instantiated Netlists

1. Slice Logic
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Site Type		Used	Fixed	Prohibited
Available	Util%			
Slice LUTs*		71	0	0
10400	0.68			
LUT as Logic		63	0	0
10400	0.61			
LUT as Memory		8	0	0
9600	0.08			
LUT as Distributed RAM		8	0	
LUT as Shift Register		0	0	
Slice Registers		35	0	0
20800	0.17			
Register as Flip Flop		35	0	0
20800	0.17			
Register as Latch		0	0	0
20800	0.00			
F7 Muxes		0	0	0
16300	0.00			
F8 Muxes		0	0	0
8150	0.00			
Unique Control Sets		6		0
8150	0.07			

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

Warning! LUT value is adjusted to account for LUT combining.

Warning! For any ECO changes, please run place_design if there are unplaced instances

** Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more information regarding control sets.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	—	—	—
0	—	—	Set
0	—	—	Reset
0	—	Set	—
0	—	Reset	—
0	Yes	—	—
0	Yes	—	Set
0	Yes	—	Reset
1	Yes	Set	—
34	Yes	Reset	—

2. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	25	0.00
RAMB36/FIFO*	0	0	0	25	0.00
RAMB18	0	0	0	50	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	45	0.00

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	26	0	0	106	24.53
Bonded IPADs	0	0	0	10	0.00
Bonded OPADs	0	0	0	4	0.00
PHY_CONTROL	0	0	0	5	0.00
PHASER_REF	0	0	0	5	0.00
OUT_FIFO	0	0	0	20	0.00
IN_FIFO	0	0	0	20	0.00
IDELAYCTRL	0	0	0	5	0.00

	IBUFDS		0		0		0	
104		0.00						
	GTPE2_CHANNEL		0		0		0	
2		0.00						
	PHASER_OUT/PHASER_OUT_PHY		0		0		0	
20		0.00						
	PHASER_IN/PHASER_IN_PHY		0		0		0	
20		0.00						
	IDELAYE2/IDELAYE2_FINEDELAY		0		0		0	
250		0.00						
	IBUFDS_GTE2		0		0		0	
2		0.00						
	ILOGIC		0		0		0	
106		0.00						
	OLOGIC		0		0		0	
106		0.00						
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5. Clocking

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	Site Type		Used		Fixed		Prohibited		Available		Util%	
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	BUFGCTRL		1		0		0		32		3.13	
	BUFIO		0		0		0		20		0.00	
	MMCME2_ADV		0		0		0		5		0.00	
	PLLE2_ADV		0		0		0		5		0.00	
	BUFMRCE		0		0		0		10		0.00	
	BUFHCE		0		0		0		72		0.00	
	BUFR		0		0		0		20		0.00	
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6. Specific Feature

Site Type	Used	Fixed	Prohibited	Available	Util%
BSCANE2	0	0	0	4	0.00
CAPTUREE2	0	0	0	1	0.00
DNA_PORT	0	0	0	1	0.00
EFUSE_USR	0	0	0	1	0.00
FRAME_ECCE2	0	0	0	1	0.00
ICAPE2	0	0	0	2	0.00
PCIE_2_1	0	0	0	1	0.00
STARTUPE2	0	0	0	1	0.00
XADC	0	0	0	1	0.00

7. Primitives

Ref Name	Used	Functional Category
FDRE	34	Flop & Latch
LUT6	25	LUT
OBUF	24	IO
LUT5	24	LUT
LUT4	10	LUT
RAMS32	8	Distributed Memory
LUT2	7	LUT
LUT3	5	LUT
CARRY4	4	CarryLogic
IBUF	2	IO
FDSE	1	Flop & Latch
BUFG	1	Clock

8. Black Boxes

+-----+	+-----+
Ref Name	Used
+-----+	+-----+

9. Instantiated Netlists

+-----+	+-----+
Ref Name	Used
+-----+	+-----+